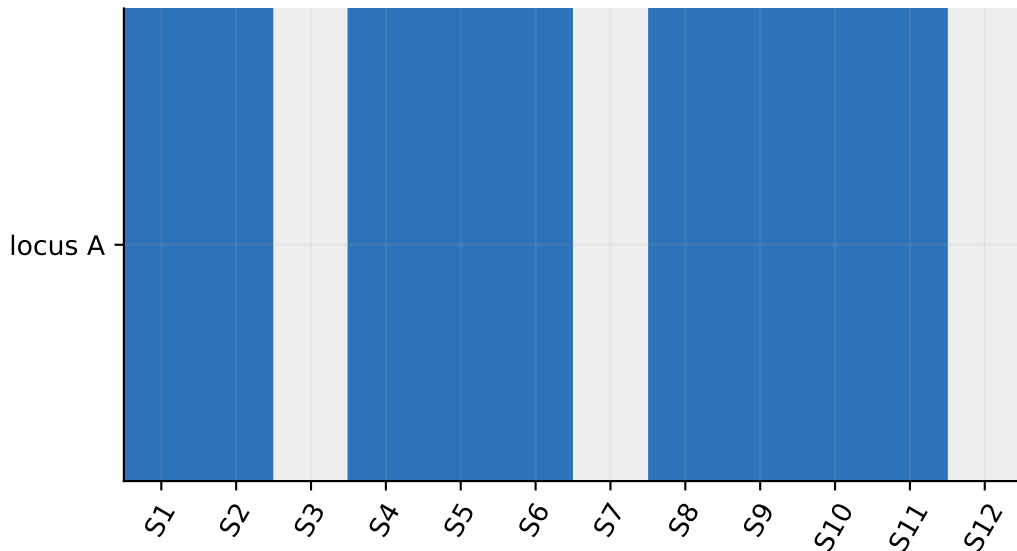
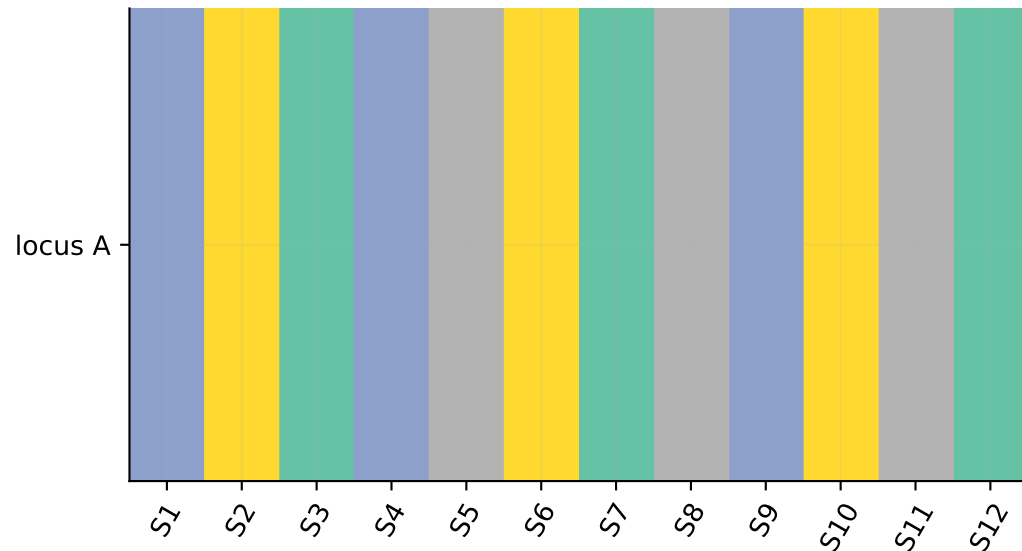


FIG02 One biological-state representation, two resolutions

Presence state $P \in \{0,1\}$



Type state $T \in \{a_1, a_2, \dots\}$



Missing/absent and nominal allele categories remain explicit; allele IDs are categories, not numerical distances.